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Association of COVID-19 outcomes with measures of institutional and interpersonal trust: an ecological analysis using national data from 61 countries

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Despite the importance of human behavior in containing a disease outbreak, formal quantitative analyses examining the relationship between measures of trust and COVID-19 outcomes remain limited. We use data from Wave 7 (2017–2022) of the World Values Survey to assess the country-level relationship between trust and COVID-19 outcomes across 61 countries via clustering and regression. After adjusting for country-level confounders, our findings indicate that countries with low trust have significantly greater numbers of COVID-19 deaths (1200.6 more COVID-19 deaths per million, 95% CI [510.92, 1890.3]), significantly greater excess death (2289.1 more excess deaths per million, 95% CI [971.1, 3607.2]), and a lower vaccination rate (16.6 fewer people vaccinated per 100, 95% CI [−27.7, −5.6]) than high trust countries, suggesting a tangible impact of trust on country-level COVID-19 outcomes. We discuss differences between interpersonal and institutional trust and advocate for incorporating trust in disease modeling to better predict country-level outcomes.

Societal trust is a well-established enhancer and facilitator of public health. Differential levels of societal trust correlate with differential public health outcomes for a range of interventions, with vaccine acceptance[1], preventative screening[2], and diagnostic testing[3] all being positively associated with increased trust in healthcare providers, science, and/or the government. The role that trust—defined as the belief and conviction that an entity will do what they say they will[4,5]—plays in compliance and adherence to public health guidelines becomes even more pronounced in acute crises such as contagious disease outbreaks, where guidance may change rapidly as new information is learned and the scale of the outbreak evolves. In 21st-century epidemics, including SARS-CoV-1, H1N1, and Ebola, higher trust in government, media, and medical organizations was linked to increased adherence to public health recommendations[6,7], decreased anxiety[8], and higher vaccine uptake[9].

This association between increased trust and improved public health outcomes has also been observed in the COVID-19 pandemic at both the individual and population levels. Higher societal trust correlated with not only better individual uptake of preventative measures[10–13] but also better compliance with country-level public health measures and better COVID-19 health outcomes. In Europe, "lockdown"-style policies designed to slow and prevent transmission reduced mobility by 14–16% more in regions with high political trust than in areas with low political trust[14] and globally, countries with over 40% of individuals reporting a high level of generalized interpersonal trust achieved a dramatic reduction in cases and deaths following spikes in COVID-19 infections[15]. Many studies have furthermore tied higher levels of institutional and interpersonal trust to lower infection fatality rates and fewer COVID-19 and excess deaths per million[16–20], as well as higher vaccine confidence[21].

The results from these studies, while affirming the importance of trust for COVID-19 health outcomes, also raise questions about the nuances of this multiplex relationship. COVID-19 outcomes may be impacted not only by varying degrees of trust, but also by different types of trust—such as interpersonal vs. institutional—that,

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due to their different underlying mechanisms, might act on metrics like mortality rates and vaccine acceptance in contrasting ways[22–27]. This is particularly the case for vaccination rate: studies on the relationship between trust and vaccination hesitancy and uptake have previously and conflictingly suggested that those hesitant to receive the COVID-19 vaccine have significantly lower levels of institutional trust than interpersonal trust[25]; that increased interpersonal trust has a negligible negative association with the probability of an individual receiving a COVID-19 vaccine[28]; and that increased interpersonal trust has a significant positive association with COVID-19 vaccination[29]. Better disentangling and understanding the tensions between institutional and interpersonal trust is crucial to optimizing public health responses to future infectious disease outbreaks and other public health emergencies. Key unanswered questions include: what types of trust, both institutional and interpersonal, are most strongly associated with COVID-19 outcomes at a country level? Do we see meaningful differences in COVID-19 outcomes across countries with varying levels of interpersonal and institutional trust?

This study aims to answer these questions by analyzing the relationship between measures of various types of trust and COVID-19 outcomes of interest. Specifically, we examine the relationships between specific types of trust—both institutional and interpersonal—and three quantitative COVID-19 health outcomes. We use trust items from Wave 7 (2017–2022) of the World Values Survey[30] and link them with three COVID-19 outcomes: confirmed COVID-19 deaths per million (2020–2022), vaccination rate of at least one dose per 100 people (2020–2023), and estimated excess deaths per million (2020–2021), sourced from publicly available data from JHU CSSE (Johns Hopkins University Center for Systems Science and Engineering)[31] and the World Health Organization (WHO)[32]. The World Values Survey (WVS) is a worldwide research program that periodically conducts cross-national, cross-sectional surveys measuring sociocultural values. All of the WVS survey data is publicly available on the WVS website[30]. The WVS data is increasingly leveraged as a resource in public health research to illuminate the intersections between social values and health outcomes[17,33–37]. Using questions from the WVS survey section "Social Capital, Trust and Organizational Membership," we form aggregated metrics that capture the countrywide level of trust in various entities, including trust in domestic government, in the WHO, in people one knows personally, and in people one meets for the first time. Additionally, we incorporate five publicly available country-level markers of development as potential confounders—gross domestic product (GDP) per capita, average life expectancy from the World Bank[38], average years of formal schooling for individuals aged 15 to 64 from the Barro-Lee dataset (processed by Our World In Data)[39], the percentage of a country's population that lives in urban areas from Our World In Data[40], and the Freedom Score democracy index from Freedom House[41]—for their potential to impact both our trust covariates and our COVID-19 outcomes.

We assess the ecological relationship between country-level trust measures and COVID-19 outcomes through regression and clustering techniques using data from 61 countries worldwide after adjusting for covariates. To evaluate which of the 23 individual trust items are most strongly associated with COVID-19 outcomes, we regress our country-level COVID-19 outcomes on each trust item one at a time, adjusting for country-level features. Then, to compare national-level COVID outcomes across these trust items jointly, we consider two-step clustering and regression methods to first assign each country to a cluster (regardless of outcome), then regress COVID-19 outcomes on countries' trust-based cluster membership. Lastly, we use a one-step method to simultaneously cluster and regress our COVID-19 outcomes on trust, where the clusters are themselves informed by the outcome of interest. Figure 1 visualizes this process and our various methodological approaches. From this analysis, we draw conclusions about the relationship between the enumerated types of trust and COVID-19 outcomes and suggest opportunities for incorporating measures of trust into public health modeling.

The rest of the paper is organized as follows. In the results section, we first describe the general structure of the individual trust items we extract from the WVS and present descriptive analyses of these items and our COVID-19 outcomes of interest. We then report our findings from our ecological regression (ordinary regression conducted at the country-level as opposed to individual survey respondent level), adjusting for country-level confounders, and identify how many individual trust items are significantly associated with COVID-19 outcomes. We then report on the two different approaches for our clustering and regression analysis (two-step and joint), demonstrating that clustering countries in terms of vectors of trust is a useful grouping for understanding COVID-19 outcomes. These two approaches allow us to both identify specific kinds of trust that are potentially informative for COVID-19 outcomes and how overall country trust profiles inform COVID-19 outcomes. We supplement our data-driven analysis with a sub-scale analysis where the trust items are thematically grouped, as opposed to data-driven clusters (see Supplementary Material). We conclude with a discussion of our findings and implications for future research, particularly the possibility of using country-level trust metrics to improve nowcasting exercises from compartmental disease transmission models. In methods that follow the results, we further detail the data extraction and summarization from the WVS and the exact analytical methods used for our analysis.

Results

Trust items

Of the 33 items in the standard questionnaire of the WVS Wave 7 that ask participants about their trust or confidence in a variety of entities, we selected 23 thematically relevant items for our research questions of interest. These 23 items capture various types of both institutional and interpersonal trust. Supplementary Table S1 contains a complete list of the 33 WVS questions and those included in our analysis. All items have ordinal response categories for the level of trust (ex. "Do not trust at all," "Trust completely") that we converted to a numerical value, with a higher number signifying greater trust. We then calculated the proportions of individuals responding in each category for each country and used these proportions and our numerical category values to create weighted country-level score metrics for each question; see Methods for further details on this process. All trust items are coded such that the lowest possible score means that all individuals in a country reported having

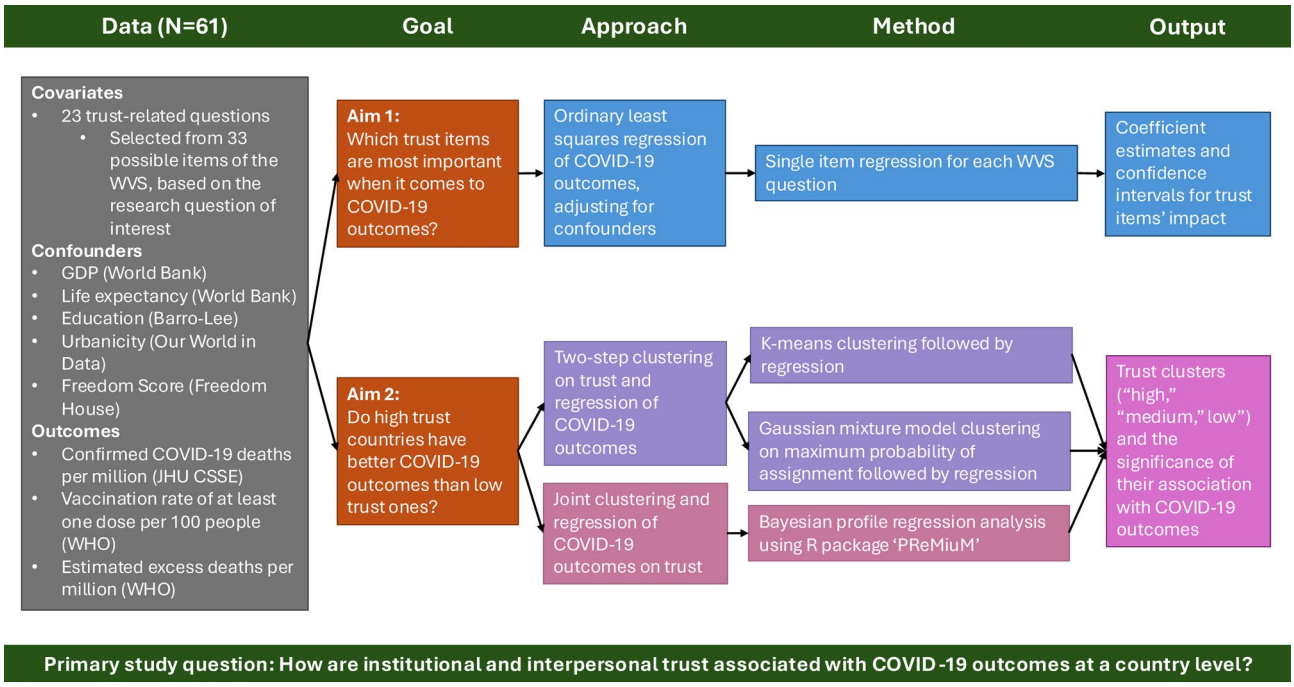

Fig. 1. A flow chart describing the data sources used for analysis, our study objective, our various study methodologies, and our analytical outputs.

the lowest level of trust in the entity questioned, and the highest possible score means that all individuals in a country reported having the highest level of trust in the entity questioned.

We conducted our primary analysis on individual trust items following a data-driven approach that would not over- or under-weight any trust item, as would happen in an analysis of sub-scales formed by combining/aggregating unequal numbers of items. However, it is a common practice to create such contextual, qualitatively driven aggregated sub-scales of trust, grouping thematically related trust items together. In order to supplement our primary, single-item analysis, we also conducted a parallel sub-scale analysis that grouped 20 of the 33 questions into seven distinct trust subdomains, based on a qualitative thematic analysis of the question topics (Supplementary Table S2.1): trust in international organizations (4 items), trust in domestic governance (6 items), trust in the WHO (1 item), trust in universities (1 item), trust in media (2 items), trust in community (3 items), and trust in strangers (3 items). Further details on how these subdomains were calculated are available in Supplementary Material Secion 2.

Descriptive statistics

Of the 66 countries participating in WVS, 61 were included in our analysis (see Methods for further details and Supplementary Figure S1 for a map of country coverage). These 61 countries include most countries in Asia and North and South America, with more limited coverage in Africa and Europe. Table 1 provides summary statistics for our covariates, country-level features, and outcomes for all 61 countries included in our analysis. The trust items with the greatest standard deviation were those asking about domestic governance (such as question 71, trust in government, and question 73, trust in parliament); these questions also had the means closest to the center of the score's range. Those with the smallest variation were those addressing interpersonal trust with "in" groups like family, neighbors, and friends; these trust items also had the highest averages. Our country-level features and outcomes of interest all demonstrate considerable spread across countries.

Our correlation analysis (see Supplementary Figure S2) of all trust items, country-level features, and outcomes suggested strong multicollinearity between trust items, particularly those concerned with institutional trust. In general, the direction of this correlation was positive: trust appears to beget trust. Within our outcomes, we observe a strong positive correlation between confirmed COVID-19 deaths per million and estimated excess deaths per million (Pearson correlation coefficient $\rho = 0.69$, p-value < 0.05), though there is low correlation between vaccination and death rates. Within our country-level features, we observe a high positive correlation between all variables (all with $\rho > 0.50$, p-value < 0.05). Despite this high positive correlation, the variance inflation factor suggested low multicollinearity (below 3 for all covariates). Our correlation analysis also identifies negative correlations between trust items and national death rates, and positive correlations between most trust items and vaccination rate.

Ecological regression

To explore how different kinds of trust impact country-level COVID-19 outcomes, we developed a series of ecological regression models for our outcomes of interest. Our analysis consisted of "single item" regression

Summary of Country-level Data ($N = 61$)	Mean (SD)
Trust item scores	
Item response range from 1 to 2	
Most people can be trusted	1.22 (0.15)
Item response range from 1 to 4	
Your family	3.74 (0.14)
Your neighborhood	2.81 (0.25)
People you know personally	2.91 (0.29)
People you meet for the first time	1.96 (0.26)
People of another religion	2.27 (0.31)
People of another nationality	2.17 (0.33)
The press	2.28 (0.32)
Television	2.37 (0.36)
Justice system/courts	2.53 (0.43)
The government	2.34 (0.49)
The political parties	2.00 (0.44)
Parliament	2.17 (0.51)
The civil services	2.41 (0.42)
Universities	2.84 (0.28)
Election	2.37 (0.43)
The environmental protection movement	2.60 (0.27)
Charitable or humanitarian organizations	2.69 (0.27)
Major regional organization (combined from country-specific regions)	2.36 (0.35)
The United Nations	2.41 (0.37)
North Atlantic Treaty Organization	2.22 (0.34)
The World Bank	2.38 (0.42)
The World Health Organization	2.69 (0.33)
Country-level Features	
GDP per capita (in $USD)	15,159.39 (17,530.36)
Life expectancy (in years)	75.29 (5.97)
Education (average years of formal education)	9.88 (2.55)
Urbanicity (% of total population living in urban areas)	66.45 (20.48)
Freedom Score (out of 100)	58.31 (28.90)
COVID-19 Outcomes	
COVID-19 deaths per million	1470.13 (1352.89)
Vaccination rate per 100 people	69.62 (19.53)
Estimated excess deaths per million	2251.69 (2243.33)

Table 1. Summary statistics (mean and standard deviation) for all trust item scores (from WVS Wave 7), country-level features (for the year 2019, sourced from the World Bank and Our World in Data), and COVID-19 outcomes (sourced from the WHO and JHU CSSE) at a country-level, for all 61 countries included in our dataset.

models—models where we regress our outcomes on one trust item, adjusting for country-level features—for all 23 questions in our analysis (Table 2). Within single item regression, adjusting for multiple testing via a Bonferroni correction, five questions (22% of all questions) were significantly negatively associated (p-value < 0.002) with COVID-19 deaths per million; six (26% of all questions) were significantly positively associated and one (4% of all questions) was significantly negatively associated with vaccination rate per 100 people; and thirteen questions (56% of all questions) were significantly negatively associated with estimated excess deaths per million.

For our subscale analysis, we conducted both "single subdomain" regression models as above and additionally explored a joint regression of all seven subdomains and confounders at once. These analyses returned similar findings and clearly highlighted the tension between interpersonal and institutional trust's impacts on COVID-19 outcomes, as discussed in the introduction. While increased "trust in community" (a measure of interpersonal trust) was associated with a decrease in excess deaths per million (−157.30 fewer deaths per million for one increased unit of trust, 95% CI [−283.19, −31.42]), it was also associated with a decrease in vaccination rate (−1.23 fewer vaccinations per 100 people for one increased unit of trust, 95% CI [−2.20, −0.26]) in our joint subdomain regression models (Supplementary Table S2.4). In comparison, increased "trust in universities" (a measure of institutional trust) was significantly associated with only desirable outcomes (fewer COVID-19 deaths per million, more vaccinations per 100 people). Complete results of the analysis at this aggregated subscale level are summarized in Supplementary Tables S2.2–4.

Question	COVID-19 deaths per million			Vaccination rate per 100 people			Estimated excess deaths per million		
	Estimate	95% Confidence Interval	*p*-value	Estimate	95% Confidence Interval	*p*-value	Estimate	95% Confidence Interval	*p*-value
Most people can be trusted	−2371.4	(−4982.5, 239.6)	0.074	24.6	(−16.9, 66.2)	0.031	−5431.3	(−10361.9, −500.8)	0.24
Your family	−1231.6	(−3647.8, 1184.5)	0.31	−45.8	(−81.9, −9.8)	0.21	−2894.0	(−7495.4, 1707.6)	0.014
Your neighborhood	−1399.1	(−2605.3, −193.0)	0.024	−13.3	(−32.7, 6.2)	0.001	−3799.1	(−5987.4, −1610.9)	0.18
People you know personally	−1175.2	(−2465.1, 114.9)	0.073	−19.0	(−39.1, 1.2)	0.022	−2843.9	(−5266.7, −421.0)	0.064
People you meet for the first time	−570.4	(−1905.3, 764.6)	0.4	−7.9	(−28.8, 13.0)	0.049	−2489.2	(−4970.2, −8.2)	0.45
People of another religion	−193.0	(−1391.9, 1006.0)	0.75	−0.4	(−19.2, 18.4)	0.53	−719.9	(−3008.4, 1568.5)	0.97
People of another nationality	−312.4	(−1619.6, 994.9)	0.63	−15.0	(−35.1, 5.1)	0.35	−1179.6	(−3666.1, 1306.9)	0.14
The press	−1662.8	(−2601.4, −724.2)	< 0.001	24.2	(9.3, 39.2)	0.0028	−2872.3	(−4707.2, −1037.4)	0.0019
Television	−1464.9	(−2363.2, −566.7)	0.0019	22.2	(8.0, 36.3)	0.0013	−2895.0	(−4602.9, −1187.1)	0.0027
Justice system/courts	−1216.1	(−1929.2, −503.1)	0.0012	9.4	(−2.6, 21.4)	< 0.001	−2568.6	(−3900.7, −1236.4)	0.12
The government	−1251.5	(−1901.0, −602.0)	< 0.001	20.6	(10.6, 30.6)	0.013	−1697.4	(−3023.2, −371.6)	< 0.001
The political parties	−1144.9	(−1891.9, −397.8)	0.0033	25.8	(15.2, 36.3)	0.0089	−1967.0	(−3420.8, −513.1)	< 0.001
Parliament	−1054.9	(−1705.1, −404.4)	0.002	22.0	(12.7, 31.4)	0.015	−1609.1	(−2897.5, −320.8)	< 0.001
The civil services	−1188.9	(−1981.1, −396.7)	0.004	16.3	(3.7, 29.0)	0.0013	−2516.8	(−4004.3, −1029.2)	0.012
Universities	−489.2	(−1619.3, 641.0)	0.39	41.7	(28.0, 55.4)	0.14	−1594.8	(−3729.0, 539.5)	< 0.001
Election	−772.1	(−1539.4, −4.9)	0.049	24.1	(13.5, 34.6)	0.035	−1573.1	(−3034.2, −112.1)	< 0.001
The environmental protection movement	−1341.1	(−2474.2, −207.9)	0.021	46.8	(33.2, 60.3)	0.012	−2800.2	(−4947.3, −653.0)	< 0.001
Charitable or humanitarian organizations	−804.8	(−2036.2, 426.6)	0.2	38.3	(21.7, 54.8)	0.0073	−3115.2	(−5353.3, −877.2)	< 0.001
Major regional organization	−1078.1	(−2171.4, 15.3)	0.053	39.3	(25.2, 53.4)	0.0025	−3150.6	(−5139.8, −1161.4)	< 0.001
The United Nations	−1113.4	(−2106.2, −120.6)	0.029	33.3	(19.9, 46.8)	< 0.001	−3201.5	(−4986.3, −1416.7)	< 0.001
North Atlantic Treaty Organization	−724.1	(−1726.9, 278.5)	0.15	26.5	(12.2, 40.8)	0.0014	−2986.5	(−4764.5, −1208.5)	< 0.001
The World Bank	−1176.6	(−2110.7, −242.6)	0.015	28.3	(15.0, 41.7)	0.0032	−2682.8	(−4425.7, −940.0)	< 0.001
The World Health Organization	−1238.2	(−2229.3, −247.1)	0.015	37.4	(24.6, 50.2)	< 0.001	−3247.9	(−5045.4, −1450.4)	< 0.001

Table 2. Results from our series of single item regression models, where each COVID-19 outcome of interest is regressed on one trust item, adjusting for GDP, life expectancy, educational attainment, urbanicity, and Freedom Score.

Two-step clustering and regression

To explore how countries may be grouped based on their levels of trust, and how these groupings may relate to COVID-19 outcomes, we sought to identify trust-based clusters and regress COVID-19 outcomes on these clusters. We first conducted a two-step clustering and regression method, where the clustering on the 23 trust items and regression of COVID-19 outcomes were done sequentially. We used two clustering methods for this two-step analysis: k-means clustering and Gaussian mixture model (GMM) classifications. Based on various statistics used for identifying the optimal number of clusters, $k = 3$ components were selected for both methods (see Methods for further details). For our GMM analysis, we used the maximum posterior probability of cluster assignment to assign countries to clusters, as there was a high degree of certainty in cluster assignment: the maximum posterior probability of cluster assignment was above 0.95 for all 61 countries.

In both methods, there was decent visual separation between clusters and balanced distribution of cluster membership across countries (see Fig. 2). There was also decent (66%) agreement between the two clustering methods in terms of country assignments, though the GMM clusters were more imbalanced than those identified by k-means, with the "medium trust" cluster containing only seven countries. To interpret these clusters, we compared the average trust subdomain scores across the k-means and GMM clusters (Table 3). Both methods divided into groups of countries with "high trust," "low trust," and "medium trust" in the trust items. The interpretation of these clusters is largely the same across these two methods: the "high trust" cluster has the highest average overall trust and the highest level of institutional trust in both methods, and "low trust" the lowest; the "medium trust" cluster, for both methods, is more ambiguous, with the highest average interpersonal trust levels but middle-of-the-pack institutional trust. In terms of the country-level features, which were not used in cluster formation, across both methods, the "medium trust" clusters have the highest average GDP, years of education, and Freedom Score; the "high trust" clusters have the lowest average urbanicity and Freedom Score; and the "low trust" cluster has the lowest average GDP. Cluster interpretation is bolstered by analyzing specific country assignment; for example, China, a country recognized for reporting high levels of overall trust[42] is assigned in both methods to the "high trust" cluster. See Supplementary Table S2 for a full list of country cluster assignments.

To assess the influence of trust cluster membership on COVID-19 outcomes, we fitted separate regressions of each outcome of interest on cluster membership indicators, adjusting for country-level features. Table 4 contains the point estimates and confidence intervals for all regression outputs, where the coefficient estimates compare

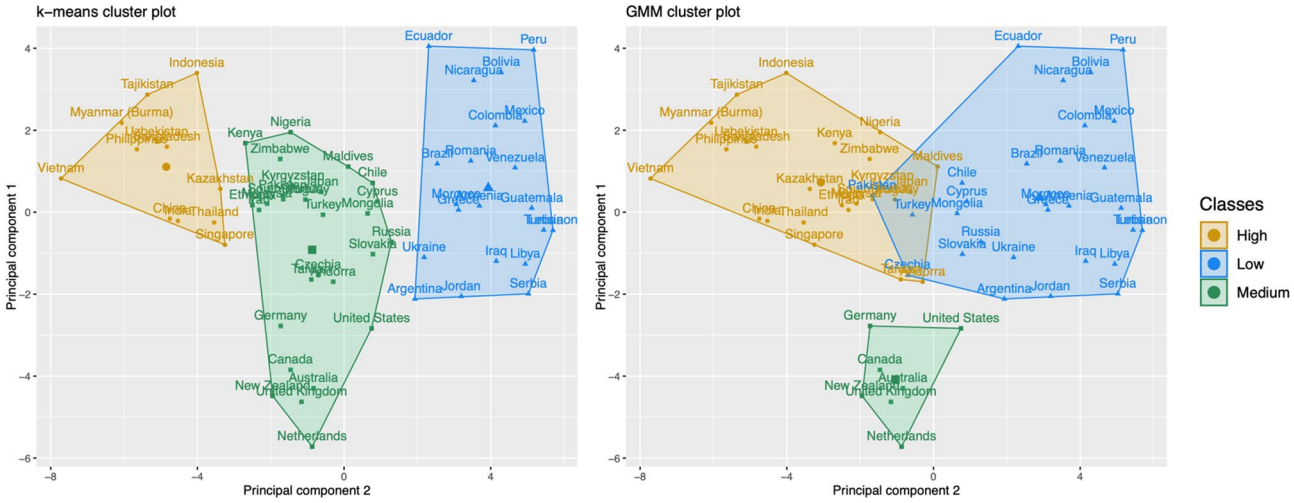

Fig. 2. Visualization of the three clusters identified by k-means (left) and GMM (right) on the first two principal components of the 23 trust items.

	k-means clusters			GMM clusters		
Question label	**Low**	**Medium**	**High**	**Low**	**Medium**	**High**
Most people can be trusted	1.12	1.28	1.24	1.15	1.51	1.22
Your family	3.69	3.73	3.83	3.70	3.72	3.79
Your neighborhood	2.62	2.89	2.93	2.70	2.99	2.88
People you know personally	2.74	3.06	2.86	2.80	3.39	2.90
People you meet for the first time	1.81	2.09	1.94	1.87	2.42	1.95
People of another religion	2.16	2.41	2.15	2.17	2.84	2.23
People of another nationality	2.03	2.34	2.04	2.06	2.86	2.11
The press	2.02	2.29	2.73	2.10	2.13	2.54
Television	2.07	2.38	2.86	2.18	2.15	2.66
Justice system/courts	2.15	2.61	3.01	2.24	2.75	2.80
The government	1.87	2.36	3.10	2.01	2.24	2.75
The political parties	1.59	2.02	2.68	1.71	1.98	2.34
Parliament	1.67	2.21	2.95	1.81	2.20	2.57
The civil services	1.95	2.51	2.97	2.10	2.47	2.75
Universities	2.59	2.89	3.14	2.67	2.87	3.02
Election	1.93	2.49	2.89	2.08	2.64	2.64
The environmental protection movement	2.35	2.66	2.90	2.40	2.63	2.83
Charitable or humanitarian organizations	2.47	2.74	2.95	2.51	2.70	2.89
Major regional organization (combined from country-specific regions)	2.01	2.44	2.77	2.10	2.33	2.66
The United Nations	2.04	2.51	2.81	2.12	2.51	2.71
North Atlantic Treaty Organization	1.94	2.31	2.48	1.99	2.51	2.40
The World Bank	2.00	2.45	2.87	2.08	2.31	2.74
The World Health Organization	2.39	2.77	3.02	2.41	2.75	2.99
Country-level features[a]						
GDP per capita (in $1000 USD)	6.99	23.75	9.42	8.98	50.18	12.52
Life expectancy (in years)	74.96	76.28	73.56	75.25	81.54	73.58
Education (average years of formal education)	9.12	10.79	9.08	9.53	13.30	9.33
Urbanicity (% of total population living in urban areas)	73.16	68.86	49.06	71.52	84.23	55.58
Freedom Score (out of 100)	55.81	70.14	35.08	59.10	95.29	47.04

Table 3. Average trust item scores and mean country-level feature values across clusters in both methods. [a] Variables not used in cluster formation.

Cluster	Estimate	95% CI[a]	*p*-value
k-means clusters			
COVID-19 deaths per million			
From high to low trust	1333.3	(375.3, 2291.3)	0.007
From high to medium trust	338.1	(−494.3, 1170.6)	0.419
Vaccination rate per 100 people			
From high to low trust	−32.0	(−45.7, −18.4)	< 0.001
From high to medium trust	−18.4	(−30.3, −6.6)	0.003
Excess deaths per million			
From high to low trust	2377.9	(592.9, 4162.9)	0.010
From high to medium trust	−88.5	(−1639.5, 1462.6)	0.909
GMM clusters			
COVID-19 deaths per million			
From high to low trust	1200.6	(510.9, 1890.3)	0.001
From high to medium trust	227.5	(−999.7, 1454.8)	0.711
Vaccination rate per 100 people			
From high to low trust	−16.6	(−27.7, −5.6)	0.004
From high to medium trust	2.4	(−17.2, 22.1)	0.805
Excess deaths per million			
From high to low trust	2289.1	(971.1, 3607.2)	0.001
From high to medium trust	−398.7	(−2744.0, 1946.7)	0.734
Outcome averages: k-means clusters[b]			
Cluster / Outcome	*Low*	*Medium*	*High*
COVID-19 deaths per million	203.9 (1463.7)	1395.9 (1214.6)	359.2 (280.1)
Vaccination rate per 100 people	62.6 (22.2)	71.3 (18.7)	78.0 (11.9)
Estimated excess deaths per million	3658.0 (2507.0)	1549.7 (1823.2)	1428.5 (1447.1)
Outcome averages: GMM clusters[b]			
Cluster / Outcome	*Low*	*Medium*	*High*
COVID-19 deaths per million	2169.7 (1451.8)	1737.6 (1145.6)	583.8 (598.4)
Vaccination rate per 100 people	64.6 (20.0)	82.7 (4.9)	71.8 (20.0)
Estimated excess deaths per million	3416.1 (2469.9)	1089.6 (1304.5)	1226.4 (1382.7)

Table 4. Results from regression on cluster membership, adjusting for country-level features, with the reference group being the "high trust" cluster, and the mean COVID-19 outcomes and standard deviations per cluster across both methods. [a]Confidence interval. [b]Results formatted as mean (standard deviation).

the mean difference in outcome in one cluster compared to our reference cluster (the "high trust" cluster), as well as the mean outcomes and standard errors for each cluster, for both k-means and GMM. For all three outcomes across both methods, there was a statistically significant difference (all p-values < 0.01) in the mean expected outcome between a "high trust" country and a "low trust" one, with the "low trust" countries having significantly worse health outcomes (more COVID-19 and excess deaths, lower vaccination rates) than the "high trust" ones. The effect size of this difference was formidable: for GMM, the membership of a "low trust" cluster was associated with 1200.6 more COVID-19 deaths per million (95% CI [510.9, 1890.3]), 2289.1 more excess deaths per million (95% CI [971.1, 3607.2]), and a 16.6 decrease of vaccination rate per 100 people (95% CI [−27.7, −5.6]) in comparison to the "high trust" cluster. Though there was always a significant difference between the "high trust" and "low trust" countries across clustering methods, the membership of a "medium trust" cluster versus a "high trust" did not result in a statistically significant difference for any of our outcomes of interest (other than vaccination rate in the k-means clusters). Similar findings were observed for our supplemental sub-scale analysis (see Supplementary Tables S2.5 and S2.6 and Supplementary Figure S2.2).

Sensitivity analysis for clustering

In all our adjusted models, we use a core set of five confounders measured for each country at a national level. To assess the robustness of our results to the choice of confounders, we expanded our initial set of five confounders by adding three more potential confounders suggested by a reviewer that may be relevant to our analysis: hospital beds per 1000 people[43], the percentage of the population over 65[44], and the government effectiveness index[45,46]. We then assessed the robustness of our results to choice of confounders we adjust for by plotting the adjusted effect estimate (adjusted mean difference in outcomes in high and low trust group for k-means clusters) across all possible 2^8 choices of covariate adjustment, sifting through 256 regression models (Supplementary Figs. 4–6). We also calculated the E-value[47] to assess the effect of unmeasured confounders. The results show that our reported estimates remain robust across adjustment choices and that the unmeasured confounder must have

a very strong effect on both COVID-outcomes and trust to negate the significant results we report as indicated by a large E-value.

Joint clustering and regression

Lastly, we carried out joint clustering and regression for each of our three outcomes of interest using a Bayesian profile regression model through Markov chain Monte Carlo. This method, where the outcomes are non-parametrically linked to our covariates through cluster membership via simultaneous clustering and regression, addresses uncertainty in cluster membership more directly than our previous methods[48]. As the outcome informs the clustering and regression in this method, we built three distinct models for our three outcomes of interest that produced three distinct cluster profiles. In order to reduce multicollinearity between the 23 trust items within the regression, only the top ten (or top five, for vaccination rate) trust items were used in each model, selected based on the single item regression p-values and effect size. The output of these models was then postprocessed to identify representative clusters that serve as a summary for all clustering explored throughout the simulations. The results of this postprocessing and clustering are in Table 5. See Methods for further details on the steps of this analysis.

In contrast to both two-step methods, our joint models identified $K = 2$ as the optimal representative clustering for the deaths per million outcomes, eliminating any kind of "medium trust" cluster, but kept a "medium trust" cluster for vaccination rate ($K = 3$ for this outcome). Most of the countries identified as "medium trust" in the two-step methods were assigned to the "high trust" cluster in the joint models, reinforcing

Question	**COVID-19 deaths per million**		**Question**	**Vaccination rate per 100 people**			**Question**	**Estimated excess deaths per million**	
Cluster	*Low trust*	*High trust*		*Low trust*	*Medium trust*	*High trust*		*Low trust*	*High trust*
Trust item scores[a]									
The press	1.11 (1.06, 1.17)	1.27 (1.23, 1.32)	The political parties	1.14 (1.04, 1.23)	1.21 (1.15, 1.27)	1.29 (1.2, 1.44)	Your neighborhood	1.07 (1.01, 1.14)	1.27 (1.23, 1.3)
Television	3.66 (3.61, 3.71)	3.79 (3.74, 3.83)	Parliament	3.66 (3.55, 3.76)	3.74 (3.69, 3.79)	3.8 (3.72, 3.91)	The press	3.62 (3.56, 3.68)	3.78 (3.75, 3.82)
Justice system/courts	2.63 (2.53, 2.72)	2.91 (2.85, 2.97)	Election	2.68 (2.53, 2.84)	2.8 (2.71, 2.9)	2.94 (2.78, 3.18)	Television	2.59 (2.49, 2.69)	2.88 (2.82, 2.94)
The government	2.76 (2.66, 2.86)	3 (2.91, 3.08)	The United Nations	2.76 (2.51, 2.95)	2.92 (2.81, 3.02)	3.02 (2.87, 3.21)	Justice system/courts	2.65 (2.5, 2.82)	3 (2.93, 3.07)
The political parties	1.78 (1.7, 1.89)	2.07 (1.99, 2.15)	The World Health Organization	1.82 (1.6, 2)	1.97 (1.88, 2.07)	2.06 (1.93, 2.22)	The civil services	1.7 (1.6, 1.79)	2.06 (2.01, 2.11)
Parliament	2.07 (1.97, 2.19)	2.39 (2.29, 2.48)					Major regional organization	2.06 (1.95, 2.16)	2.35 (2.26, 2.43)
The civil services	1.9 (1.8, 2.01)	2.33 (2.25, 2.4)					The United Nations	1.91 (1.77, 2.05)	2.27 (2.18, 2.35)
The environmental protection movement	2.49 (2.32, 2.71)	3.02 (2.86, 3.17)					North Atlantic Treaty Organization	2.41 (2.22, 2.61)	2.98 (2.85, 3.1)
The World Bank	2.56 (2.38, 2.78)	3.03 (2.89, 3.16)					The World Bank	2.54 (2.39, 2.7)	2.97 (2.85, 3.09)
The World Health Organization	2.08 (1.94, 2.23)	2.4 (2.3, 2.49)					The World Health Organization	2.05 (1.95, 2.17)	2.36 (1.28, 2.45)
Cluster information									
Average outcome	2248.04	1031.31	Average outcome	44.91	74.42	77.08	Average outcome	4470.34	1462.84
GDP	7.47	19.5	GDP	6.34	19.51	9.47	GDP	6.58	18.21
Life expectancy	74.92	75.5	Life expectancy	74.68	75.9	73.93	Life expectancy	74.78	75.47
Education	9.48	10.11	Education	8.91	10.36	9.26	Education	9.11	10.16
Urbanicity	72.1	63.26	Urbanicity	68.89	70.06	52.76	Urbanicity	70.49	65.01
Freedom Score (out of 100)	55.09	60.13	Freedom Score	49.36	69.37	31.5	Freedom Score	55.19	59.42
Number of countries	39	22	Number of countries	11	38	12	Number of countries	16	45
Average silhouette width	0.80		Average silhouette width	0.59			Average silhouette width	0.86	

Table 5. Results from the representative clusters identified in postprocessing of our bayesian profile regression models. [a]Results are reported with the posterior mean and the 95% highest posterior density intervals. Each outcome has its own model. GDP is reported in 1000 $USD; life expectancy is reported in years; education in average years of formal education; urbanicity in the percentage of total population living in urban areas; and the Freedom Score is out of 100.

the idea that the largest gap is between "high trust" and "low trust" countries (see Supplementary Table S2). The average silhouette width, which evaluates clusters by measuring how similar a subject is to other subjects in its cluster[49] was above 0.8 for the deaths per million outcomes, suggesting strong cluster cohesion and separation. However, the average silhouette width was only 0.59 for vaccination rate, suggesting there is more ambiguity between clusters in this model.

Similar to the two-step clustering, the "high trust" cluster countries have higher average trust item scores than those in the "low trust" cluster. These differences are more pronounced for the deaths per million models, where there is very little overlap between the highest posterior density intervals of each cluster's average trust item scores. As seen in the two-step methods, these "high trust" and "low trust" clusters had highly different mean outcomes across all three models. Similar findings were observed for our supplemental sub-scale analysis (see Supplemental Table S2.7).

Discussion

Our analysis reveals several important nuances of the relationship between trust and COVID-19 and supports the use of societal trust metrics in COVID-19 modeling. Our study has several key strengths. To our knowledge, no other country-level analysis focusing on both different types of trust and multiple COVID-19 outcomes exists in extant literature. By leveraging the WVS data, we achieve a cross-national evaluation of how both institutional and interpersonal trust relate to COVID-19 outcomes at a country level. We examine the relationship between trust and estimated excess deaths, a COVID-19 outcome that has been understudied in this context, despite excess deaths often being a more reliable measurement of COVID-19-related mortality due to its accounting of underreporting and cause of death misclassification[50].

While the majority of the single item regressions suggested that increased trust was associated with better health outcomes—lower deaths, higher vaccination rates—this was not uniformly observed. While increased trust, both institutional and interpersonal, was always associated with a decrease in both confirmed COVID-19 and estimated excess deaths per million, this was not the case for vaccination rate. Though all of the statistically significant institutional trust items were positively associated with vaccination rate, the one statistically significant interpersonal trust item, trust in one's neighborhood, was negatively associated, i.e. greater trust in this setting was associated with a decrease in vaccination rate. As discussed in the introduction, existing research on the direction of interpersonal trust's influence on vaccination rate—a conceptually distinct outcome from deaths per million[51,52]—has yielded conflicting results. Our analysis provides further evidence of a negative association between interpersonal community trust—particularly trust in one's neighborhood—and vaccine uptake. It is possible that high community trust correlates with cultural resistance to institutional authorities, as people prioritize local community structures over centralized governance[53,54]. It may also reflect a misplaced trust that one's neighbors and friends would not infect each other. This high community trust may reduce willingness to get vaccinated, as seen in our analysis.

Both our two-step and joint clustering analyses found that trust-based cluster membership was statistically significant for all outcomes of interest, particularly between the "high trust" and "low trust" clusters. When it came to cluster formation, institutional trust items played the most significant role in differentiating each cluster. In particular, the most extreme across-cluster differences in trust appeared in domestic governance (the government, political parties, and elections). Average interpersonal trust levels were less important and less variable across clusters. Notably, however, they did not align with the trend we observed in the institutional trust items. In both two-step methods, the "high trust" clusters did not have the highest levels of interpersonal trust—instead, the "medium trust" clusters had the highest averages, and "high trust" and "low trust" interpersonal levels were often comparable. Here again, we see that institutional and interpersonal trust do not necessarily act in tandem. Our results suggest further research is needed to fully untangle the ways interpersonal trust influences different public health outcomes, separate from institutional trust.

Across our analyses, there was a clear difference in COVID-19 outcomes between "high trust" and "low trust" countries. The distinction between "high trust" and "medium trust" was more ambiguous, indicating there may not be a monotone positive ordering in this relationship. Only k-means clustering identified a significant difference between "high trust" and "medium trust" countries, and only for one of our three outcomes of interest: vaccination rate. This ambiguity also appears in our joint clustering and regression approach. When we included country-level features and outcome of interest in our clustering model, there was only one outcome, vaccination rate, for which including a "medium trust" cluster was chosen for inclusion. While one potential explanation for this ambiguous "medium trust" cluster is potential nonlinearity to the relationship between trust and COVID-19 outcomes, we believe the explanation for this likely lies in differences between country-level features/covariates. In the two-step approach and the joint clustering approach for vaccination rate, the "medium trust" countries differ significantly in terms of country-level features: these countries have a markedly higher per capita GDP and a higher Freedom Score than countries in either the "low trust" or "high trust" clusters (see Tables 3 and 5). These differences, particularly the greater material resources afforded by a higher GDP nation, could be effectively closing the gap between "medium trust" and "high trust" countries when it comes to COVID-19 outcomes—perhaps a relative surplus of wealth makes up for a relative dearth of trust. The mediating effect of a country's wealth and infrastructure on the relationship between trust and health outcomes suggested by our findings offers an interesting avenue for further research. Yet regardless of this potential effect, our analysis demonstrates that material resources alone do not determine positive public health outcomes. Despite their more comparable per capita GDP and Freedom Score, the "low trust" and "high trust" clusters have significantly disparate COVID-19 outcomes in both two-step approaches. Trust does matter, and in ways we can quantitatively capture.

These clear associations between trust and COVID-19 outcomes—even when using conceptually distinct approaches—open the door to the possibility of meaningfully incorporating these trust metrics into disease transmission modeling. Many models now incorporate various time-dependent and time-invariant elements of

human behavior for more accurate estimations of disease transmission[55–57]. This offers many potential pathways for country-level trust metrics' inclusion. For example, trust metrics could be included as coefficients in the regression models, such as MERMAID[58], used to estimate the rate parameters for compartmental models. These parameter estimates are increasingly based not only on specific properties of the pathogen (like duration of infectiousness) but also on external factors, such as government policy or emotional contagion[59]. Our study suggests that trust, including institutional trust, has the potential to be meaningfully included as external, constant effect metrics in such models.

Our study has several limitations. As the WVS is an ecological study, we must beware the ecological fallacy of treating these across-country results as indicative of within-country trends. WVS Wave 7 was interrupted by the COVID-19 pandemic, leading to about a dozen countries' surveys being conducted after the pandemic had begun—an event which could have influenced the responses of individuals in those countries. The WVS also had limited coverage in Africa and Europe. Many European countries participate in the European Values Study[60], a survey done in conjunction with the WVS but without several of the key questions, such as trust in the WHO and universities, used in our analysis. This more limited global coverage restricts the generalizability of our findings and their applicability to certain regions. Lastly, our outcome of estimated excess deaths per million, taken from the WHO, was estimated with large uncertainty intervals[50]. Our study does not consider the impact of this uncertainty on our findings.

Institutional and interpersonal trust have played crucial roles in the unfolding of the COVID-19 pandemic. By considering the impact of trust in disease transmission models, prioritizing a better understanding of interpersonal trust's influence on public health outcomes, and attaining high levels of trust in science and public health, we can further improve our responses to disease outbreaks and save lives.

Methods

Data sources

We used data from the WVS Wave 7 as our source for all trust-related covariates. The WVS Wave 7 is the seventh iteration of the cross-national, cross-sectional social survey periodically conducted by the World Values Survey, a non-commercial research program. For Wave 7, over 90,000 adult respondents across 66 countries and regions were interviewed (primarily via face-to-face interviews, with some phone interviews for remote regions) with a standardized questionnaire covering social, ethical, religious, and economic values. Respondents were selected via random probability representative samples within each country surveyed, with a minimum sample size of 1,000 for all countries. Data collection began in mid-2017 and, due to COVID-19-related delays, was completed in July 2023[30]. Further details of how we analyzed the WVS dataset are below.

Our primary COVID-19 outcomes of interest were confirmed COVID-19 deaths per million from January 1, 2020 to December 31, 2022, sourced from JHU CSSE COVID-19 Data[31], the vaccination rate of at least one dose per 100 people from 2020 to 2023, and excess deaths per million in 2020 and 2021, the latter both sourced from the WHO COVID-19 Surveillance Database[32,50,61].

Our confounding variables, GDP per capita (in current $USD) and total life expectancy (in years), were sourced from the World Bank[38]; educational attainment (average years of schooling for individuals aged 15–64) was sourced from the Barro-Lee dataset, processed by Our World In Data[39]; urbanicity (percentage of the country's population living in urban areas) was sourced from Our World In Data[40]; Freedom Score, a measure of democracy, was sourced from Freedom House[41]. The 2019 metrics were used to capture the status of each country when the COVID-19 pandemic first began.

This covariate, outcome, and country-level feature data resulted in a complete dataset for 61 countries of the 66 surveyed by WVS. For Taiwan, different data sources were necessary for outcome data: its excess deaths estimate was sourced from the Economist[62]. Missing GDP per capita and life expectancy data from the World Bank were alternatively sourced from the International Monetary Fund[63] and the CIA World Factbook[64]; educational attainment was sourced from a report from Taiwan's Ministry of Education[65]; and urbanicity was sourced from Worldometer[66]. Four surveyed regions (Hong Kong, Macau, Puerto Rico, and Northern Ireland) are accounted for in their country-level outcome data (China, the United States, and the United Kingdom, respectively). The final excluded country was Egypt, due to the high levels of nonresponse for WVS survey questions.

Defining and extracting trust covariates from WVS wave 7

We sought to define trust based on the questions in the WVS Wave 7 survey that included the key words "trust" or "confidence" and were part of the standard questionnaire given to all participating countries. In social science theory, confidence and trust are widely recognized as adjacent, though not equivalent, concepts. Confidence is the belief that an entity will do what they are expected to do and trust is a thornier belief made up not only of confidence in an entity but also faith in an entity's integrity and competence[4,5]. In WVS Wave 7, "trust" is used exclusively when asking about interpersonal relations and "confidence" is used exclusively when asking about institutions. We will use the term "trust" to refer to all of the WVS questions, with "interpersonal trust" corresponding to the questions actually labeled "trust" in WVS and "institutional trust" corresponding to the questions actually labeled "confidence." This decision is motivated by the similarity between the definitions of these two concepts and existing research based on the WVS that also uses the term "trust" to refer to both items labeled "trust" and "confidence"[17].

Thirty-three questions met these criteria, coded $t = 57 \ldots 89$, all from the "Social Capital, Trust and Organizational Membership" subsection of the survey. Thirty-two of the 33 questions followed a parallel response structure of four ordinal and four nominal responses, while one question (Q57, "Most people can be trusted") had two ordinal and four nominal response categories. These three data structures, alongside an example of what the data looks like for one country's individual responses to one question, are shown in Fig. 3.

Question on "Trust": Q57 ("Most people can be trusted")	
Non-response	*Response*
-1: "Don't know"	1: "Most people can be trusted"
-2: "No answer	2: "Need to be very careful"
-4: "Not asked"	
-5: "Missing/NA"	
Questions on "Trust": Q58–63	
Non-response	*Response*
-1: "Don't know"	1: "Trust completely"
-2: "No answer"	2: "Trust somewhat"
-4: "Not asked"	3: "Do not trust very much"
-5: "Missing/NA"	4: "Do not trust at all"
Questions on "Confidence": Q64–89	
Non-response	*Response*
-1: "Don't know"	1: "A great deal"
-2: "No answer"	2: "Quite a lot"
-4: "Not asked"	3: "Not very much"
-5: "Missing/Unknown"	4: "None at all"

Question 71: "How much confidence do you have in the government?" for the United States

Response	Cases	Percentage
"A great deal"	190	7.3%
"Quite a lot"	610	23.5%
"Don't know/No answer"	17	0.7%
"Not very much"	986	38.0%
"None at all"	793	30.5%
"Not available/Missing"	0	0.0%
Total	2596	100%

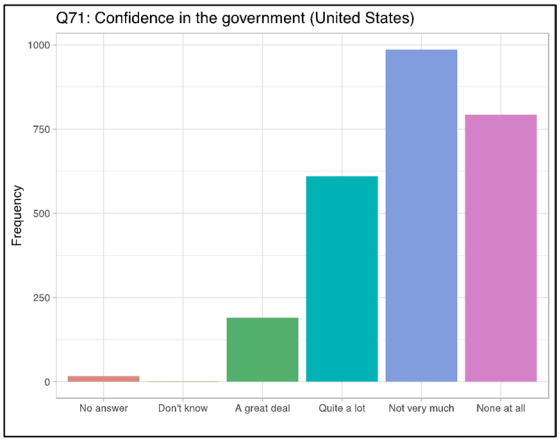

Fig. 3. The ordinal and nominal response options for our survey questions of interest, alongside an example of this data for the United States and the question "How much confidence do you have in the government?"

Not all of the 33 eligible questions from the WVS were deemed relevant to our research question of interest, primarily because the mechanism through which trust in this entity would act upon our COVID-19 outcomes appeared conceptually ambiguous, such as with trust in the women's movement (question 80). These thematically irrelevant questions run the risk of introducing non-informative noise to our model, as they do not pertain to our research aim. In order to clearly operationalize our research goal and remove these spurious questions, we conducted a qualitative analysis of the 33 available items. This process identified key thematic groups through which trust could act upon COVID-19 outcomes: trust in "in" interpersonal groups, such as one's family; trust in "out" interpersonal groups, such as those of another nationality; trust in media, such as the press; trust in bodies of science, such as universities; trust in domestic governance, such as parliament; and trust in international governance, such as the United Nations. These thematic groups were used to identify questions to include in our analysis; some other individual questions were also selected based on prior literature suggesting a correlation between trust in these entities and trust in public health and medicine. For example, environmental protection is frequently classified as part of public health[67] leading to the inclusion of question 79 ("Confidence: The Environmental Protection Movement"). In the end, 23 of the 33 questions were selected for our analysis. See Supplementary Table S1 for a full list of the available WVS questions and those used in our analysis.

Each country ($j = 1 \ldots 61$) in Wave 7 has some $n_{j,t}$ individual responses to each question t. As our outcomes of interest are at a population level, we sought to aggregate our trust item data from the individual level to the population level so that each country had one data point for each item. For our aggregation, we excluded the nominal non-responses and focused on the ordinal categorical data.

The total *n* given by the WVS dataset includes counts of non-responses. For our sample proportions, we will use an adjusted *n*:

$$n^{*}_{j,t} = n - (n_{-1,j,t} + n_{-2,j,t} + n_{-4,j,t} + n_{-5,j,t})$$

where $n_{i,j,t}$ with $i = [-5, -4, -2, -1, 1, 2, 3, 4]$ corresponds to the possible non-response or response categories for country at question t. This adjusted $n^{*}_{j,t}$ thus accounts only for response values.

Each country's count data $(n_{1,j,t}, n_{2,j,t}, n_{3,j,t}, n_{4,j,t})$ follow a multinomial distribution with parameters $n^{*}_{j,t}$ and $(\pi_1, \pi_2, \pi_3, \pi_4)$ where $n^{*}_{j,t} = \Sigma_k n_{k,j,t}$ is the fixed number of responses (we assume each individual response is independent and identically distributed) and there are $k = 4$ categories of response, such that π_k is the probability of observing response k and $\Sigma_k \pi_k = 1$. The multinomial log-likelihood is:

$$L(\pi_{j,t}) = \Sigma_k n_{k,j,t} \log(\pi)_{k,j,t}$$

Our maximum likelihood estimates are the sample proportions, calculated as $p^{k}_{j,t} = \frac{n_{k,j,t}}{n^{*}_{j,t}}$ and

$n^*_{j,t} = \Sigma_k n_{k,j,t}$. Thus

$$\sum_{k=1}^{4} p^k_{j,t} = \sum_{k=1}^{4} \frac{n_{k,j,t}}{n^*_{j,t}} = 1.$$

Using these sample proportions, we can calculate a weighted question score as follows:

$$TS_{j,t} = p^1_{j,t}*4 + p^2_{j,t}*3 + p^3_{j,t}*2 + p^4_{j,t}*1$$

where the weights are the inverse of the recorded structure such that the higher the score, the higher the level of trust. This procedure is used for all t questions with a four-category ordinal structure, regardless of the wording of the responses, such that the "trust" and "confidence" questions are comparable. Question 57 was similarly aggregated according to its two-category structure, with p_1 as 2 ("Most people can be trusted") and p_2 as 1 ("Need to be very careful"):$TS_{j,57} = p^1_{j,57}*2 + p^2_{j,57}*1$

This resulted then a 61×23 matrix of countries and question scores used for analysis.

There is substantial within-question variation of country trust scores, particularly for questions about both domestic and international governmental organizations (see Supplementary Figure S3). The most stable questions are those about charities and interpersonal trust, particularly trust in members of one's community or family. Our exploratory analysis also revealed a series of complexities around missing data in the survey. Country-level non-response was rather low: only a small number of questions were entirely omitted from surveys in a small number of countries. However, individual item non-response, where an individual was asked a question but chose not to respond, varied considerably. While mean individual non-response was below 10% across countries for over three-quarters of the 23 questions, this was not always the case. This was seen particularly in Egypt, where the mean non-response was over 30% and went as high as 74.7%. This led to Egypt's responses being excluded from further analysis. No other countries were excluded for missingness, and the few remaining question scores with non-response rates above 40% were replaced via single imputation with the median score from countries with missingness below 40%.

Statistical analysis: single item regression

To understand how individual trust items were associated with COVID-19 outcomes, we conducted ordinary least-squares regression. The associations between trust and our COVID-19 outcomes of interest were assessed by a series of single item regression models, adjusting for our five country-level features. For each trust item and each outcome of interest, the following model was used:

$$\mathrm{E}(Y_r) = \beta_{0,r} + \beta_{1,r}\mathrm{X} + \beta_{2,r}\mathrm{GDP} + \beta_{3,r} Life\ expectancy + \beta_{4,r} Education$$
$$+\beta_{5,r} Urbanicity + \beta_{6,r} Freedom\ Score$$

where the subscript *r* denotes that the model is built with respect to the *r*-th COVID-19 outcome of interest (confirmed deaths per million, COVID-19 vaccination rate per 100 people, estimated excess deaths per million), and *X* denotes the question of interest (any of the 23 items included in our analysis).

None of the outcomes were transformed for regression. Measures of GDP, life expectancy, education, urbanicity, and democracy were included to control for potential confounding from healthcare infrastructure, population health, and overall country development. While a multiple linear regression model was considered, the high levels of collinearity between the trust items, coupled with the relatively low sample size ($N = 61$) made even sparse regression techniques like elastic net regression unstable and ultimately uninformative.

Statistical analysis: deterministic and probabilistic trust clustering and regression

We sought to classify countries based on their level of trust and analyze whether cluster membership impacted COVID-19 outcomes (i.e. whether being "high trust" was associated with better COVID-19 outcomes than being a "low trust" country). We used a two-step process that first identified clusters based on trust item scores and then regressed our COVID-19 outcomes of interest on cluster membership, adjusting for our country-level features. The first clustering approach used was k-means clustering, which minimizes within-cluster variances for a pre-specified number of k clusters. A variety of visual and analytical tools—including assessing the AIC (Akaike Information criterion) and BIC (Bayesian information criterion), gap statistics, and elbow method results—were used to select $k = 3$ clusters based on this algorithm.

The second two-stage approach was clustering using finite Gaussian mixture modeling. Here, we assume that the datapoints for each country are generated by a distribution made up of k groups and k corresponding components. This distribution is of the form

$$\mathrm{f}(\mathrm{x_i}; \Psi) = \Sigma_{k=1}^{K} \pi_k f_k(x_i; \beta_k).$$

where $\psi = (\pi_1 \ldots \pi_{K-1}, \beta_1 \ldots \beta_k)$ is the vector of all model parameters, $f_k(x_i; \beta_k)$ is the k-th component's density function for x_i with parameters β_k, and π_k is the mixing probability of belonging to component k such that $\Sigma_{k=1}^{K} \pi_k = 1$. The data's log-likelihood is then

$$logL(\Psi) = \Sigma_{k=1}^{K} \Sigma_{j=1}^{n} z_{kj} (log\pi_k + logf_k(x_i; \beta_k)).$$

where z_{kj} is the indicator for whether y_j belongs to component k. As we assume each component's density is Gaussian, we can write $f_k(x_i; \beta_k) \sim \mathrm{N}(\mu_k, \Sigma_k)$. The expectation-maximization algorithm can then be used to obtain estimates for μ_k, Σ_k, and π_k. This is the approach taken in the R package 'mclust', which was used for this analysis[68]. Through an analysis of BIC, ICL (integrated completed likelihood), and log-likelihood values, $k = 3$ components were selected.

For both methods, ordinary least-squares linear regression was used to assess the statistical significance of cluster membership in relation to our three COVID-19 outcomes of interest and trust subdomain scores. Our outcomes were individually regressed on a categorical variable for cluster membership, adjusting for country-level features, as shown below:

$$\begin{aligned}\mathrm{E}(Y_r) = {} & \beta_{0,r} + \beta_{1,r} I(C_i = Low) + \beta_{2,r} I(C_i = Medium) \\ & + \beta_{3,r} GDP + \beta_{4,r} Life\ expectancy + \beta_{5,r} Education + \beta_{6,r} Urbanicity + \beta_{7,r} Freedom\ Score\end{aligned}$$

with the subscript r denoting that the model is built with respect to the r-th COVID-19 outcome of interest (confirmed deaths per million, excess deaths per million, COVID-19 vaccination rate per 100 people), C_i referring to cluster membership of country i, and $\beta_{0,r}$ representing the "high trust" cluster coefficient.

Statistical analysis: Bayesian profile regression

After our two-step clustering, we sought to employ a method that could perform simultaneous clustering and regression and thus better incorporate uncertainty in cluster membership into the model. To do this, we used Bayesian profile regression as implemented in the R package 'PReMiuM'[69]. Based on Dirichlet process mixture modeling (DPMM), this framework uses Markov Chain Monte Carlo methods to non-parametrically link the outcome and covariates through cluster membership while adjusting for country-level features[48,70]. Though this DPMM-based method is not a direct joint analog to our two-step methods (which use finite mixture models for GMM), we chose to leverage this gold standard method to be able to preserve the continuous nature of our covariates. As the outcome variable is considered in the clustering produced by this method, we ran three distinct joint profile regression models for each of our COVID-19 outcomes of interest, so that, in contrast to our two-step findings, we had different clustering assignments for each outcome.

For each COVID-19 outcome of interest y_i, our model is $y_i = \theta_{z_i} + \beta^T W_i + \epsilon_i$, where θ_{z_i} is the random effect coefficient of cluster membership, $\beta^T W_i$ is the term encapsulating the fixed effects (our five country-level features) whose coefficients β do not change with cluster membership, and ϵ_i is the normally distributed error term. As both our covariates (the trust item scores) and outcomes are continuous, we assumed a mixture of Gaussian distributions for covariates and a Gaussian distribution for our response y_i, whose mean μ_i is represented by $\mu_i = \theta_{z_i} + \beta^T W_i$. A blocked Gibbs sampler was then used to generate estimates from h runs. From these runs, partition around medoids on the dissimilarity matrix was used to identify the 'best' partitions for a list of possible number of clusters, which was then maximized via average silhouette width to identify an overall 'best' and representative partition. For further details, consult the 'PReMiuM' manual[69].

To run the models for each outcome, we scaled and centered all covariates and country-level features. Because the number of parameters (23 trust items and five country-level features) was large relative to our sample size, we chose a subset of the trust items to use in our analysis to mitigate multicollinearity. We selected these subsets by ranking the single item regression outputs by p-value and magnitude of association to identify the most important trust items for each outcome. For the two deaths per million outcomes, the top ten questions were used; for our vaccination rate, only the top five were used to guarantee a stable model. The distribution of vaccination rate across countries—already more ambiguous than death rates when clustered by trust, as seen in our two-step analysis—likely contributed to the difficulty in finding a stable model for this outcome-influenced regression. Once the questions were selected, we used a burn-in period of $h = 5000$ runs with 10,000 recorded sweeps, default priors provided by 'PReMiuM', and an initial number of clusters as two. This clustering was not fixed and changed throughout the model runs; the final output for the two death rates identified the optimal number of clusters as two, but the final output for vaccination rate identified the optimal number as three. Postprocessing returned representative clusters summarizing all clustering explored throughout the algorithm. Using these representative clusters, we calculated highest posterior density intervals for all trust subdomain scores and the empirical mean outcomes for each cluster. Finally, we back-transformed these results from their scaling and centering for ease of comparison with our two-step findings.

Data availability

All data and code used for this analysis can be found on GitHub: https://github.com/lirountree/trustandcovid.

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LR wrote the main manuscript text and conducted all data analyses. BM conceptualized the manuscript. GT, FB, and MR provided intellectual guidance. All authors reviewed and edited the manuscript.

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